

Exam: Power Analysis

Multiple Choice (Single Select) - 4 Answer Options

1. What does power (test strength) mean?

- A) Effect size
- B) Sample size
- C) Probability of Type I error
- D) Probability of detecting an effect as significant

2. What power is conventionally required?

- A) > 0.50
- B) > 0.90
- C) > 0.80
- D) > 0.95

3. What is Type II error (beta)?

- A) Retaining H_0 when H_1 is true
- B) Sample is too small
- C) Rejecting H_0 when H_0 is true
- D) Test is significant

4. What is a priori power analysis?

- A) Calculation after the study
- B) Post-hoc power
- C) Effect size estimation
- D) Planning sample size before the study

5. Why is post-hoc power problematic?

- A) It is too accurate
- B) It cannot be computed
- C) It requires large samples
- D) It is a 1:1 transformation of the p-value

6. Which software is standard for power analysis?

- A) Word
- B) SPSS
- C) G*Power
- D) Excel

7. Which effect size is used for t-tests?

- A) Cohen's d
- B) Cramer's V
- C) Eta-squared
- D) R-squared

8. What happens to power with smaller sample size?

- A) Alpha error increases
- B) Power stays the same
- C) Power increases
- D) Power decreases

Solutions

1: D

2: C

3: A

4: D

5: D

6: C

7: A

8: D